

9.0 DUPLEX MODE CONFIGURATION AND NEGOTIATION

The ENC28J60 does not support automatic duplex negotiation. If it is connected to an automatic duplex negotiation enabled network switch or Ethernet controller, the ENC28J60 will be detected as a half-duplex device. To communicate in Full-Duplex mode, the ENC28J60 and the remote node (switch, router or Ethernet controller) must be manually configured for full-duplex operation.

9.1 Half-Duplex Operation

The ENC28J60 operates in Half-Duplex mode when $\text{MACON3.FULDPX} = 0$ and $\text{PHCON1.PDPXMD} = 0$. If only one of these two bits is set, the ENC28J60 will be in an indeterminate state and not function correctly. Since switching between Full and Half-Duplex modes may result in this indeterminate state, the host controller should not transmit any packets (maintain ECON1.TXRTS clear) and packet reception should be disabled (ECON1.RXEN and ESTAT.RXBUSY should be clear) during this period.

In Half-Duplex mode, only one Ethernet controller may be transmitting on the physical medium at any time. If the host controller sets the ECON1.TXRTS bit, requesting that a packet be transmitted while another Ethernet controller is already transmitting, the ENC28J60 will delay, waiting for the remote transmitter to stop. After the transmission stops, the ENC28J60 will attempt to transmit its packet. If another Ethernet controller starts transmitting at approximately the same time that the ENC28J60 starts transmitting, the data on the wire will become corrupt and a collision will occur. The hardware will handle this condition in one of two ways:

1. If the collision occurs before the number of bytes specified by the "Collision Window" in MACLCON2 were transmitted, the ECON1.TXRTS bit will remain set, a random exponential back off delay will elapse as defined by the IEEE 802.3 specification and then a new attempt to transmit the packet from the beginning will occur. The host controller will not need to intervene. If the number of retransmission attempts already matches the "Retransmission Maximum" (RETMAX) defined in MACLCON1 , the packet will be aborted and ECON1.TXRTS will be cleared. The host controller will then be responsible for taking appropriate action. The host controller will be able to determine that the packet was aborted instead of being successfully transmitted by reading the ESTAT.TXABRT flag. For more information, see **Section 7.1 "Transmitting Packets"**. A transmit abort will cause the transmit error interrupt.

2. If the collision occurs after the number of bytes specified by the "Collision Window" in MACLCON2 were transmitted, the packet will be immediately aborted without any retransmission attempts. Ordinarily, in IEEE 802.3 compliant networks which are properly configured, this late collision will not occur. User intervention may be required to correct the issue. This problem may occur as a result of a full-duplex node attempting to transmit on the half-duplex medium. Alternately, the ENC28J60 may be attempting to operate in Half-Duplex mode while it may be connected to a full-duplex network. Excessively long cabling and network size may also be a possible cause of late collisions.

When set in Half-Duplex mode, the Reset default configuration will loop transmitted packets back to itself. Unless the receive filter configuration filters these packets out, they will be written into the circular receive buffer, just as any other network traffic. To stop this behavior, the host controller should set the PHCON2.HDLDIS bit.

9.2 Full-Duplex Operation

The ENC28J60 operates in Full-Duplex mode when $\text{MACON3.FULDPX} = 1$ and $\text{PHCON1.PDPXMD} = 1$. If only one of these two bits is clear, the ENC28J60 will be in an indeterminate state and not function correctly. Since switching between Full and Half-Duplex modes may result in this indeterminate state, the host controller should not transmit any packets (maintain ECON1.TXRTS clear) and packet reception should be disabled (ECON1.RXEN and ESTAT.RXBUSY should be clear) during this period.

In Full-Duplex mode, packets will be transmitted simultaneously while packets may be received. Given this, it is impossible to cause any collisions when transmitting packets. Several configuration fields, such as "Retransmission Maximum" (RETMAX) in MACLCON1 and "Collision Window" (COLWIN) in MACLCON2 , will not be used.

When set in Full-Duplex mode, the Reset default configuration will not loop transmitted packets back to itself. If loopback is desired for diagnostic purposes, the PHCON1.PLOOPBK bit should be set by the host controller. Enabling loopback in Full-Duplex mode will disable the twisted-pair output driver and ignore all incoming data, thus dropping any link (if established). All packets received as a result of the loopback configuration will be subject to all enabled receive filters, just as ordinary network traffic would be.

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NOTES: